KAIZEN MATHS

Lesson starter

Trigonometric Differentiation

Core Trig Derivatives

Name _____

S1

Differentiate with respect to x.

$y = 2\tan(3x + 1)$

S2

Differentiate with respect to x.

$y = (\cos(-3x + 4))^2$

S3

Differentiate with respect to x.

$y = \csc(-7x + 4)$

S4

Differentiate with respect to x.

$y = 3\tan(7x + 5)$

Show your working

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ALG-C0E7-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Trigonometric Differentiation

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Name _____

E1

Differentiate with respect to x.

$y = (\sec(5x - 7))^3$

E2

Differentiate with respect to x.

$y = \csc(5x - 2)$

E3

Differentiate with respect to x.

$y = 3\sin(9x + 5)$

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ALG-C0E7-L1 · E

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ALG-C0E7-L1 · E

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Differentiate with respect to x.

$y = 3\sin(9x + 5)$

ALG-C0E7-L1 · E

Show your working

Teacher answers & working

Trigonometric Differentiation · Core Trig Derivatives

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S1

Differentiate with respect to x.

$y = 2\tan(3x+1)$

Answer:

$\frac{dy}{dx} = 6\ \sec^2(3x+1)$

Working:

Identify the inner function: $u = 3x+1$.

Differentiate the inner function: $\frac{du}{dx} = 3$.

Differentiate the outer trig function: $\frac{d}{du}(\tan(u)) = \sec^2(u)$.

Apply the chain rule: $\frac{dy}{dx} = 2\left(\sec^2(3x+1)\right)(3)$.

Multiply the constants and tidy the sign: $\frac{dy}{dx} = 6\ \sec^2(3x+1)$

S2

Differentiate with respect to x.

$y = (\cos(-3x+4))^2$

Answer:

$\frac{dy}{dx} = 6\,(\cos(-3x+4))^1\ \sin(-3x+4)$

Working:

There are three layers: the power, the trig function, and the linear inner function.

Let $v = \cos(-3x+4)$, so $y = v^2$ and $\frac{dy}{dv} = 2v^1$.

Now differentiate $v = \cos(-3x+4)$: $\frac{dv}{dx} = (-\sin(-3x+4))(-3)$.

$y' = 2(\cos(-3x+4))^1\,(-\sin(-3x+4))(-3)$.

$\frac{dy}{dx} = 6\,(\cos(-3x+4))^1\ \sin(-3x+4)$

Teacher answers & working

Trigonometric Differentiation · Core Trig Derivatives

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S3

Differentiate with respect to x.

$y = \csc(-7x + 4)$

Answer:

$\frac{dy}{dx} = 7\ \csc(-7x + 4)\cot(-7x + 4)$

Working:

Let $u = -7x + 4$, so $u' = -7$.

Recall the derivative rule: $\frac{d}{du}\csc(u) = -\csc(u)\cot(u)$.

Apply the chain rule by multiplying by u'.

$y' = (-\csc(-7x + 4)\cot(-7x + 4))(-7)$.

Therefore $\frac{dy}{dx} = 7\ \csc(-7x + 4)\cot(-7x + 4)$

S4

Differentiate with respect to x.

$y = 3\tan(7x + 5)$

Answer:

$\frac{dy}{dx} = 21\ \sec^2(7x + 5)$

Working:

Identify the inner function: $u = 7x + 5$.

Differentiate the inner function: $\frac{du}{dx} = 7$.

Differentiate the outer trig function: $\frac{d}{du}(\tan(u)) = \sec^2(u)$.

Apply the chain rule: $\frac{dy}{dx} = 3\left(\sec^2(7x + 5)\right)(7)$.

Multiply the constants and tidy the sign: $\frac{dy}{dx} = 21\ \sec^2(7x + 5)$

Teacher answers & working

Trigonometric Differentiation · Core Trig Derivatives

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E1

Differentiate with respect to x.

$y = (\sec(5x - 7))^3$

Answer:

$\frac{dy}{dx} = 15\,(\sec(5x - 7))^2\,\sec(5x - 7)\tan(5x - 7)$

Working:

There are three layers: the power, the trig function, and the linear inner function.

Let $v = \sec(5x - 7)$, so $y = v^3$ and $\frac{dy}{dv} = 3v^2$.

Now differentiate $v = \sec(5x - 7)$: $\frac{dv}{dx} = (\sec(5x - 7)\tan(5x - 7))(5)$.

$y' = 3(\sec(5x - 7))^2\,(\sec(5x - 7)\tan(5x - 7))(5)$.

$\frac{dy}{dx} = 15\,(\sec(5x - 7))^2\,\sec(5x - 7)\tan(5x - 7)$

E2

Differentiate with respect to x.

$y = \csc(5x - 2)$

Answer:

$\frac{dy}{dx} = -5\,\csc(5x - 2)\cot(5x - 2)$

Working:

Let $u = 5x - 2$, so $u' = 5$.

Recall the derivative rule: $\frac{d}{du}\csc(u) = -\csc(u)\cot(u)$.

Apply the chain rule by multiplying by u'.

$y' = (-\csc(5x - 2)\cot(5x - 2))(5)$.

Therefore $\frac{dy}{dx} = -5\,\csc(5x - 2)\cot(5x - 2)$

Teacher answers & working

Trigonometric Differentiation · Core Trig Derivatives

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E3

Differentiate with respect to x.

$y = 3\sin(9x + 5)$

Answer:

$\frac{dy}{dx} = 27\ \cos(9x + 5)$

Working:

Identify the inner function: $u = 9x + 5$.

Differentiate the inner function: $\frac{du}{dx} = 9$.

Differentiate the outer trig function: $\frac{d}{du}(\sin(u)) = \cos(u)$.

Apply the chain rule: $\frac{dy}{dx} = 3\,(\cos(9x + 5))(9)$.

Multiply the constants and tidy the sign: $\frac{dy}{dx} = 27\ \cos(9x + 5)$